Computer Vision Final Project

The final project should focus on a problem in computer vision or a closely related area in which visual processing is a central component. Each project may be completed individually or in a group of up to three students.

1 Schedule

1. **By September 30: Team Formation**

   Complete team formation on iSpace. Each project may be completed individually or in a group of up to three students.

2. **By September 30: Self-Proposed Topic Registration**

   Students who choose a self-proposed topic must send an email to wentaocheng@bnbu.edu.cn. The email should include a brief description of the proposed topic (approximately 300–500 words), explaining the problem, main idea, and planned project scope.

3. **By October 9: Topic Approval**

   Feedback on the eligibility and suitability of self-proposed topics will be provided by October 9.

4. **By December 15: Final Submission**

   Submit the final project materials through iSpace by December 15. **Late submissions will not be accepted.**

2 Topic Overview

Students may choose from Track 1, Track 2, or propose their own topic.

- *Track 1: Course-Oriented Projects*

  Track 1 projects are closely connected to the methods covered in class. Students are expected to start from a classical or course-related computer vision pipeline and extend it with learning-based approaches when appropriate.

- *Track 2: Research-Oriented Projects*

  Track 2 projects are based on recent computer vision research papers and public implementations. Students are expected to start from an existing research work, understand its main idea and implementation, and then explore a concrete research question, extension, or lightweight modification.

  All provided Track 2 topics have been carefully selected to be feasible on a single consumer GPU. An RTX 4090-class GPU should be sufficient for the proposed experiments, while many experiments can also be conducted on an RTX 4060 with appropriate model sizes and input settings.

  Track 2 is high-risk, high-return. A strong project may provide a useful foundation for a future FYP or further research toward a potential publication.

- *Self-Proposed Topics*

  Self-proposed topics are welcome as long as computer vision is a central component of the project. The topic should have a clear problem definition, a feasible implementation plan, and a reasonable way to evaluate the results.

3 Final Submission

The minimum required submission is a PDF project report. Students are strongly encouraged to prepare the report using LaTeX. The NeurIPS 2026 paper format can be used as a reference: NeurIPS 2026 Call for Papers and Paper Template.

Using the NeurIPS-style format as a reference, the final report should contain at least **4 pages of main content**, excluding references.

A working demo, demo video, or short recorded presentation is encouraged, especially for projects with interactive or visually demonstrable results.

4 Evaluation

1. *Track 1: Course-Oriented Projects*

   Evaluation will focus on the completeness and usability of the overall system. Individual modules should be correct and verifiable, and intermediate results should be clearly presented whenever possible. Meaningful integration or comparison between classical and learning-based approaches is also expected.

2. *Track 2: Research-Oriented Projects*

   Evaluation will focus on the depth of the research question, the completeness of the experiments, and the quality of the analysis. Students should identify potential weaknesses or failure cases and propose a meaningful extension. Simple hyperparameter tuning is not sufficient.

   Performance improvement is not required for Track 2. A well-motivated method with careful experiments and analysis can receive a strong evaluation even if it does not outperform the baseline.

5 Topic List

5.1 Track 1

- **From Matches to Panoramas: Image Stitching with RANSAC**

  Build an image stitching pipeline based on local feature detection, feature matching, RANSAC-based homography estimation, and image warping. Students may further explore learned feature descriptors or matchers and study how they behave under large viewpoint changes, illumination changes, or difficult image overlap.

- **From Histograms to DINO: Image Retrieval Across Representations**

  Develop a KNN-based image retrieval system using different image representations. Students may begin with simple color histograms and handcrafted features, and then extend the system with CNN, ViT, or DINO features to study how visual representation affects retrieval quality and robustness.

- **Beyond Clean Images: Robust Object Detection**

  Study how an object detector behaves when input images are degraded by blur, noise, low illumination, compression, or occlusion. Students may use a lightweight detector such as YOLO and investigate robustness, failure cases, and possible improvements through augmentation or preprocessing.

- **Beyond Filters: Classical and Learning-Based Image Denoising**

  Compare classical image denoising methods, such as Gaussian, median, or bilateral filtering, with learning-based approaches such as DnCNN. Experiments should consider different noise levels or noise types and analyze the trade-off between noise removal and preservation of image details.

- **From Interpolation to Learning: Image Super-Resolution**

  Study single-image super-resolution starting from classical interpolation methods such as bicubic interpolation and extending to learning-based approaches such as SRCNN, EDSR, or SwinIR. Students may investigate different upscaling factors, image structures, and reconstruction quality.

5.2 Track 2

- **Training-Free Multi-Prompt Fusion for Open-Vocabulary Detection**

  Starting from YOLOE, investigate whether multiple text prompts or visual exemplars can provide more reliable open-vocabulary detection than a single prompt. Students may design training-free fusion, weighting, prompt selection, or confidence-calibration strategies and evaluate them under challenging visual conditions.

- **Self-Calibrating Monocular Geometry at Test Time**

  Starting from MoGe, explore whether multiple transformed views of the same image can provide useful consistency constraints for camera estimation. Students may design a training-free self-calibration strategy that refines focal length or field of view and then improves depth or 3D geometry prediction.

- **Adaptive Memory for Training-Free Streaming Video Depth**

  Starting from Video Depth Anything, study how temporal memory or cached features should be updated during streaming depth estimation. Students may develop adaptive memory-refresh strategies based on motion, scene changes, confidence, or temporal consistency to improve long-video robustness.

- **Adaptive Visual Token Pruning for Vision-Language Models**

  Starting from VisionZip, investigate whether different image-question pairs should use different visual-token budgets. Students may design question-aware or content-aware token selection strategies to improve the trade-off between accuracy, memory consumption, and inference speed.

- **Adaptive Object-Centric Head Selection in Frozen DINO Features**

  Starting from Object-DINO, explore how object-centric information is distributed across DINO layers, attention heads, and Q/K/V representations. Students may develop image-adaptive head-selection or feature-fusion strategies for unsupervised object discovery while keeping the method training-free.